

Joining of metal-plastic hybrid composites using pre-mixture by laser assisted additive manufacturing process

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ABSTRACT

Recent studies on metal-plastic bonding have led to growing interest in hybrid 3D printing and joining materials. Until now, most hybrids have focused on printing plastic on metal, excluding metal on plastic. Printing metal on plastic has not been achieved due to significant carbonization of the bottom plastic layer caused by the thermal penetrating of the laser. This research aims to solve this issue by placing a pre-mixture of plastic and metal on the plastic layer. The existing method of laser sintering Ti6Al4V metal on polyethylene terephthalate glycol-modified (PETG) caused mutual permeation between the plastic and metal powders, resulting in a degradation in interfacial adhesion. In contrast, the pre-mixture formed an intermediate interface, preventing the interfacial permeation of powders and suppressing pore formation between the plastic and metal. As a result, the porosity analysis using X-ray tomography shows that the pre-mixture interface effectively suppresses the number of pores between PETG and Ti6Al4V, which is comparatively less than that of the interface without the mixture. This indicates that the pre-mixture strengthens the bond between the plastic and metal while protecting the interface from laser-induced damage.

1. Introduction

The field of product manufacturing is transitioning from the utilization of a single material type (e.g., metals or plastics) to the combination of diverse materials for product assembly. This shift is extensively exemplified by the use of metal-polymer for aircraft and medical implants [1,2], metal-fiber for wearable electronics [3,4], fiber-polymer for eco-friendly materials [5], and high strength materials [6–8]. In particular, cutting-edge convergence of technologies such as electronic circuits [9] and microelectromechanical systems (MEMS) [10] that feature both conductivity and insulation requires the simultaneous use of metals and plastics. In the future automotive industry as well, the development of metal-plastic composites is emerging as a major research topic to meet the increased demand for multifunctional lightweight materials.

Over the past decade, various methods for bonding metal and plastic have been developed across all industries, including the following typical methods: riveting [11], mechanical & structural fastening [12],

friction stir welding [13,14]. Notably, several approaches have emerged utilizing different materials, such as adhesive bonding with organic solvents [15], structural fastening based on injection molding [16], friction lap welding driven by friction heat [17], and the direct laser joining technique known as laser-assisted metal and plastic (LAMP) [18]. These innovations reflect a wide range of techniques designed to enhance the bonding of dissimilar materials. Furthermore, much effort has been recently made to go beyond these traditional joining techniques, eventually presenting the emergence of hybrid 3D printing technology, which provides a way to create three-dimensional structures by building up metal and plastic powders. As shown in Fig. 1a, previous our research has focused on plastic printing on metal and exploring the mechanical interlocking of metal layers [2]. Printing metal on plastic, on the other hand, remains a challenge due to the heterogeneity of processing conditions for both materials, which hinders the homogeneous formation of a cohesive plastic-metal intermediate [1]. This issue has been roughly addressed by a simple method of printing metal on the electroless-plated plastic layer, which was introduced as multi-material

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digital light processing 3D printing (MM-DLP3DP) [19] and fused deposition modeling (FDM) with catalytic precursors [20]. However, these methods still have unsolved works to overcome low productivity and limited ability to process low-melting-point metals.

The emerging high-energy laser sintering has been employed for joining high-melting-point metals in multi-material bonding, metal-on-plastic lamination [1,2], and metal-to-metal bonding [21]. The intense thermal energy generated during laser sintering, however, can expose the underlying plastic layer to thermal damage. This excessive thermal exposure results in the formation of a carbonized layer, as shown in Fig. 1b, where the plastic does not endure the heat accumulation and undergoes charring. Therefore, research aimed at reducing plastic carbonization due to laser-induced heat accumulation becomes crucial. This approach can be applied to commercialized carbon fiber reinforced polymer (CFRP) composites and lead to the development of a hybrid CFRP-metal bonding technology [22] using thermoplastic PET having a melting temperature of 220 °C and a decomposition temperature of 260 °C as the matrix polymer.

In this study, we present a novel strategy to achieve strong plastic-metal bonding while reducing plastic thermal damage by depositing a pre-mixture intermediate consisting of Ti6Al-4 V and polyvinyl alcohol (PVA). The printing of Ti-6Al-4 V alloy powders on polyethylene terephthalate glycol-modified (PETG) was performed by selective laser sintering (SLS) [23–25]. Scanning electron microscopy (SEM), energy dispersive spectroscopy (EDS), and X-ray microscopy (XRM) were used to compare printed plastic-metal laminations with and without the pre-mixture. As a result, the structural relationship between the plastic and metal was quantitatively analyzed, and the bonding mechanism, damage accumulation, porosity, and metal permeation rate associated with the pre-mixture intermediate were elucidated. This research aimed to design the strong homogeneous interfacial bond between plastic and metal by inserting pre-mixed powder as the intermediate material [2] compared to direct laser 3D printing of plastic-metal hybrids showing a heterogeneous plastic-metal intermediate. From a bonding technology perspective, this material combination can be applied to lightweight structures such as dashboards for Urban Air Mobility (UAM) vehicles

and chemically resistant industrial pipes where plastic components are embedded within metal tubing. Furthermore, the high biocompatibility of Ti and the flexible outer shell properties of PET also make this combination suitable for implantable medical structures and wearable healthcare devices.

2. Experimental methods

2.1. 3D printing of plastic substrate

To perform a feasibility test for metal printing on plastic, a plastic substrate of PETG (Shenzhen eSUN Industrial Co., Ltd., China) was prepared and fixed on a build plate before metal printing. The size of the plastic substrate is $100 \times 100 \times 2 \text{ mm}^3$ and it was printed using the fused deposition modeling (FDM) process. During printing, the nozzle temperature was 220 °C and the bed temperature was maintained at 80 °C. The printing speed was 50 mm/s, and the travel speed without extrusion was 70 mm/s. The single-layer thickness was 0.2 mm, and a total of 10 layers were printed with a 100 % infill rotation pattern.

2.2. Deposition process of pre-mixture intermediate

Selective Laser Sintering (SLS) was used to print pre-mixture intermediates and plastic-metal hybrids in succession. This 3D printer, specifically designed for plastic-metal hybrid deposition, is equipped with a build plate featuring a working area of $200 \text{ mm} \times 200 \text{ mm} \times 200 \text{ mm}$. It incorporates two material supply units: one for metal powder and the other for applying the mixture. Additionally, the system is outfitted with fiber and CO₂ laser modules, all enclosed within a sealed chamber. The laser module, positioned at the top of the chamber, is controlled through a single integrated software. The material layers on the build plate are sintered and melted by focusing the laser beam. Hybrid plastic-metal samples are manufactured following the process. The metal alloy powder used in this study was Ti6Al4V obtained by gas atomization (PAC Co., USA) with a spherical shape and size of 45–150 μm. The chemical composition of the Ti6Al4V powder is presented in Table 1.

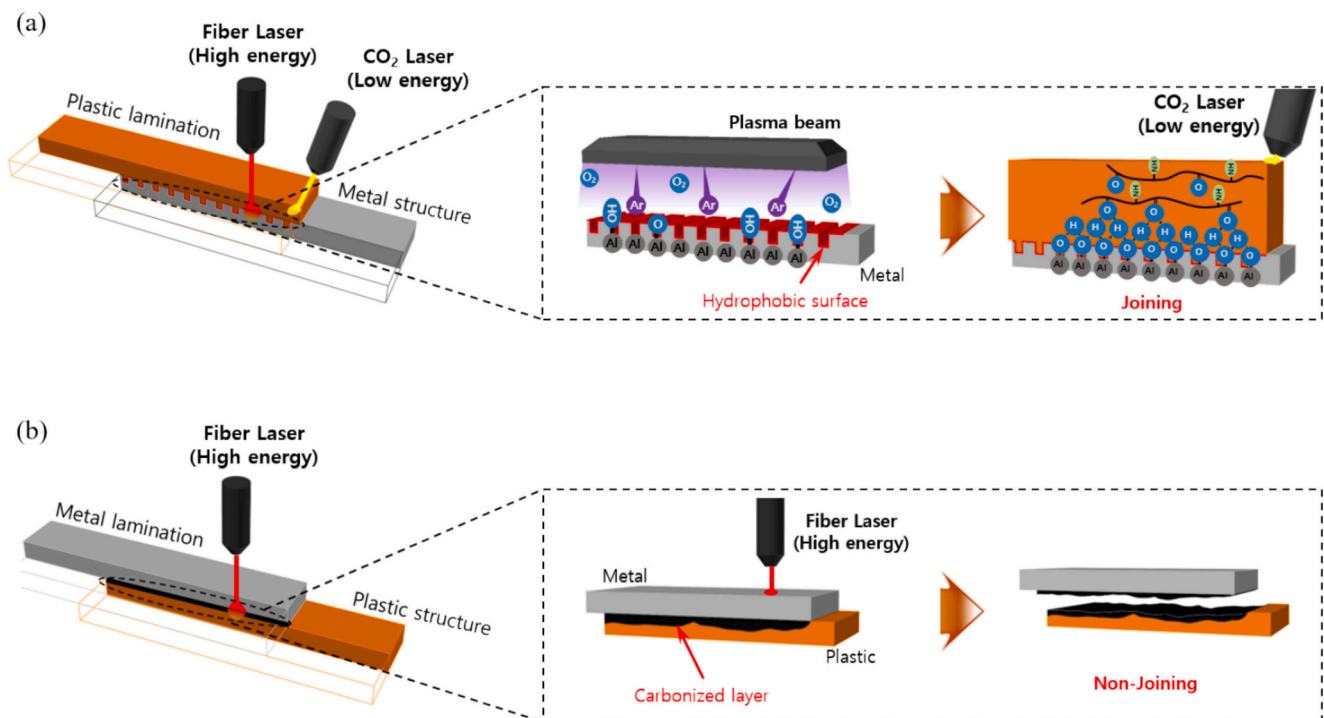


Fig. 1. Schematic illustration showing the metal-plastic joining methods and the laser sintering issues: (a) Plastic laser printing on metal [2], (b) Metal laser printing on plastic.

Table 1
Chemical compositions of the Ti6Al4V powder (PAC Co., USA).

ALLOYS	Ti	Al	V	Fe	O	C	N	H ₂	Res Total
Ti6Al4V	Bal	5.5–5.6	3.5–4.5	0.25max	0.20max	0.08	0.03max	0.0125max	0.4

The plastic component of the mixture was prepared by dissolving polyvinyl alcohol (PVA, Shenzhen Esun Industrial Co., Ltd., China) in deionized (DI) water as a solvent. Ti-6Al4V metal powder (30 g) and PVA (9 g) were mixed in DI water (50 ml, 90 °C, 1 h) and deposited onto the plastic layer fixed on the build plate using a recoater. During the recoater process, the Ti-6Al4V-PVA mixture was spread approximately 0.3 mm thick. The deposited layer was dried in a vacuum oven at 60 °C until completely dewatered. The sample was then sintered using sequential dual laser processes to produce the mixture intermediate layer. Here, dual lasers with different wavelengths were employed, a CO₂ laser with a wavelength of 10.7 μm was used to solidify position Ti6Al4V in the fluidly molten PVA, and a fiber laser with a wavelength of 1.06 μm was used to densify the molten PVA while maintaining the dispersion state of Ti6Al4V. The thickness of the mixture (~300 μm) was based on the metal permeation depth during metal laser printing.

The specifications and the detailed processing conditions of the lasers used for the experiments are detailed in Tables 2, 3, and 4. To ensure strong fixation of Ti6Al4V to PVA, a CO₂ laser [2] with a laser spot size of 250 μm and a wave length of 10.7 μm was utilized. The laser sintering process was performed with a hatch spacing of 0.2 mm and a layer height of 0.3 mm, employing a laser speed of 200 mm/s, a laser power of 12.5 W, a duty cycle of 10 %, and a frequency of 25 kHz. To further enhance the density of the mixture, it was subsequently irradiated with a fiber laser and sintered using the same hatch spacing of 0.2 mm, with a laser power of 175 W and a laser speed of 900 mm/s. For the metal printing process, a fiber laser with a laser spot size of 180 μm and a wave length of 1.06 μm was utilized. 6 layers were printed, each with a hatch spacing of 0.2 mm and a layer height of 0.05 mm. Both the contour and volume were scanned using a laser power of 175 W and a laser speed of 900 mm/s.

2.3. Laser 3D printing of metal on a plastic substrate

In this printing, the powder-bed laser sintering [2] using a fiber laser with a wavelength of 1.07 μm was employed, which involves spreading powder material with a recoater blade before sintering it layer-by-layer using a laser (Fig. 2). Prior to printing, the Ti6Al4V powder was heated in an oven at 100 °C for 12 h to eliminate remaining moisture, followed by filtration through a mesh to obtain uniformly sized particles. The prepared Ti6Al4V alloy powder was then loaded into the material supply part. Subsequently, the bed was horizontally adjusted by setting a gap of 0.05 mm, corresponding to the thickness of a single layer, between the recoater and the bed. For Ti6Al4V metal printing on each PETG plastic substrate, the layer height was set to 300 μm, with the laser’s maximum power and speed set at 175 W and 900 mm/s, respectively. The direction of sintering was alternated vertically with each layer to facilitate the lamination process. In addition, inert argon (Ar) gas was supplied at a flow rate of 50 slm during metal sintering to eliminate remaining oxygen. Once the O₂ concentration in the chamber was reduced to 1 %, the flow rate of Ar gas was reduced to 10 slm, and the laser was prepared for operation. The operation controller was

activated to initiate the laser, while the cooling system circulated to regulate the temperature during the laser process. When sintering metal on plastic, the degree of formation changes depending on the laser output value. According to the study by Yuan-Hui Chueh [1], when laminating Cu powder on PA plastic, the layer thickness (t) of 0.04 mm and the hatching space (h) of 0.2 mm were fixed while varying the laser power (P) within the range of 50–55 W and the velocity (v) set to 150–300 mm/s. Based on the data from this research, the energy density required for sintering metal (Ti-6Al4V) onto plastic (PETG) was referenced. Subsequently, metal formability was compared by calculating the volumetric energy density (E_v; J/mm³) [26] for Ti6Al4V powder laminated onto PETG plastic, using Eq. (1) [27,28]. In particular, the study by Snehashis Pal [27] measured the minimum energy density of Ti-6Al4V as 39 J/mm³, which was used as a basis for sintering metal onto plastic.

$$E_v = \frac{P}{v \times h \times t} \tag{1}$$

2.4. Plastic-metal bonding microstructure analysis

For cross-sectional image analysis of the interface between plastic and metal, a high-resolution Scanning Electron Microscope (SEM) of FEI Verios 460 (FEI Co., USA) was used. Energy Dispersive Spectroscopy (EDS, EDAX Inc., USA) was used for elemental mapping at a magnification of 100×, and the elements of the interest were mapped at the point level (PETG-Ti6Al4V: 204800 pixels, PETG-Mixture-Ti6Al4V: 204770 pixels) to compare elemental distributions. Additionally, to measure the three-dimensional interface between plastic and metal, Xradia 520 Versa (Carl Zeiss, Germany) of X-ray microscopy (XRM) was employed. This advanced analytical tool utilizes a two-stage magnification approach, enabling submicron resolution. The computed tomography process follows a series of steps, including data treatment and image acquisition. The laminated sample was positioned between the X-ray source and detector, with the X-ray beam aimed at the area of interest. The volume of interest was then scanned through a 360-degree rotation. The measurement was conducted under specific conditions, including camera binning set to 2, source settings of 40 kV and 3 W, using a 4× magnification objective and ACT (Absorption Contrast Tomography) mode. With an exposure time of 10 s per projection image, a total of 1601 images were captured at a resolution of 2.6 μm/voxel. The raw images were processed using commercial software from Carl Zeiss AG to create 3D reconstructed images. Finally, GeoDict (Math2Market GmbH, Germany) was used to process CT data, designate the pore and solid regions, and perform 3D image-based simulations. In GeoDict segmentation mode, AI-based segmentation has a training function to enhance the ability to designate pore and solid regions. The internal porosity was analyzed through PoroDict, and the porosity images specified by pore size were visualized in three dimensions.

Table 2
Fiber laser beam specification (IPG Photonics Corp., USA).

Fiber laser (YLR-500-MM-AC-TR-Y11 2011 Series)	Nominal output power (W)	Emission wavelength (nm)	Emission linewidth (nm)	Short-term power instability (rms %)	Long-term power instability (%)	Delivery fiber length (m)	Beam quality (mm × mrad)		
							Option 1–50 μm core fiber	Option 2–100 μm core fiber	Option 3–200 μm core fiber
	450–500	1070	4–5	1.0–2.0	±1– ± 3	5.0	1–2	2–5	5–15

Table 3
CO₂ laser beam specification (Coherent Corp., USA).

CO ₂ laser (DIAMOND C-55 L)	Wavelength (μm)	Optical output power (W)	Power stability (%)	Mode quality (M ²)	Beam size (mm)	Beam divergence (mrad, full angle)	Polarization	Operating frequency and duty cycle
	10.55–10.63	55	±3	<1.2	1.8 ± 0.2	7.5 ± 0.5	>100 to 1	0–25 kHz, 0–100%DC

Table 4
Processing conditions for pre-mixture intermediate and plastic-metal hybrids.

	Plastic printing	Mixture deposition & printing			Metal printing
Method	FDM printing	Recoater	CO ₂ laser	Fiber laser	Ti6Al4V
Material	PETG (Filament)	Ti6Al4V + PVA (Mixture)	Ti6Al4V + PVA (Solid phase)		: Fiber laser Ti6Al4V (Powder)
Layer height	0.2 mm × 10 layer (2 t)	0.3 mm (0.3 t)	0.3 mm × 1 layer (0.3 t)	0.05 mm × 1 layer (0.05 t)	0.05 mm × 6 layer (0.3 t)
Hatch spacing	Fused deposition	NA	0.2 mm	0.2 mm	0.2 mm
Laser speed	(220 °C, 50 mm/s)		200 mm/s	900 mm/s	900 mm/s
Laser power			12.5 W	175 W	175 W
Energy density			6.3 J/mm ³	19.4 J/mm ³	19.4 J/mm ³

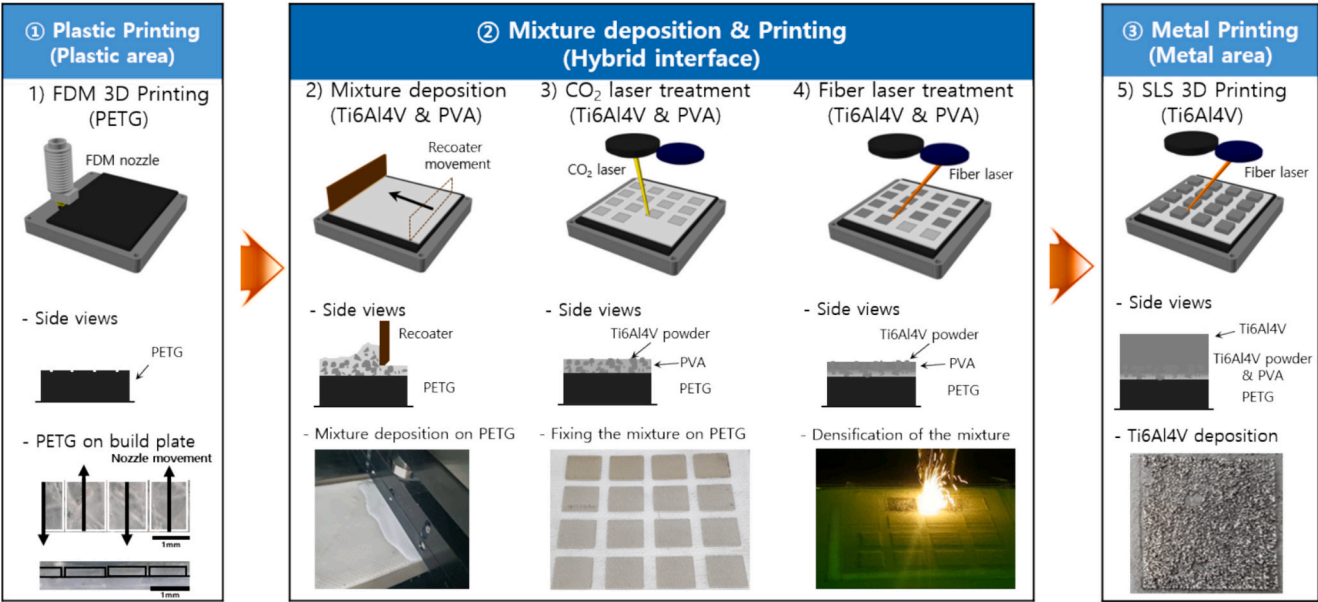


Fig. 2. Schematic diagram of the mixture formation process for metal printing on plastic.

2.5. Measuring the tensile-shear strength between plastic-metal bonding

The interfacial bonding strength of the 3D-printed hybrid specimens was evaluated using the method described in a previous study [2]. The ST-1001 Universal Testing Machine (SALT Co., South Korea), equipped with a 30 kN load cell, was used to apply the load required for debonding the hybrid specimens. Tensile testing was performed on specimens mounted on a mechanically interlocked structure at a constant cross-head speed of 0.5 mm/min. The interfacial bonding strength was determined by calculating the bonded area based on the tensile results recorded at final failure.

3. Results and discussion

3.1. Metal laser printing on a mixture-deposited plastic substrate

For strong plastic-metal bonding, the formation of a plastic-metal interface with excellent compatibility and affinity for the involved materials is a key factor. It is believed that a mixture of plastic and metal

can act as a cohesive interface between the metal and the plastic while simultaneously reinforcing the thermal capability of the plastic layer. To achieve this, it is important that the mixture is uniformly deposited on the plastic surface without affecting the underlying structure of PETG. PETG has good chemical resistance [29], but it can be damaged by certain solvents, so careful selection of the solvent used in the mixture should be ensured. In this regard, a water-soluble PVA was prepared as the plastic component of the mixture. The PVA was dissolved in deionized water (DI) and mixed with Ti6Al4V metal powder to prepare a mixture solution that could be deposited onto a plastic layer. Then, the laser affords adhesion to the mixture, assigning it to behave as an intermediate interface. The laser treatment was performed in two steps using two different wavelengths of laser beams. As shown in Fig. 2, the CO₂ laser with a wavelength of 10.5 μm has high polymer absorption and low metal absorption, and these characteristics were used to solidly position the Ti6Al4V powders in the molten PVA. Next, the fiber laser with a wavelength of 1.07 μm densified the molten PVA while maintaining the dispersion state of the Ti6Al4V powders. As a reference, our previous studies reported the underlying mechanism by which infrared

lasers of two different wavelengths assigned adhesive functions at the plastic-metal interface.

Subsequently, the optimal irradiation conditions for the fiber laser should be determined to print Ti6Al4V metal powder on the intermediate interface. As shown at Fig. 3(a), the total 25 samples using 5×5 array were deposited using different laser power and laser speed. The energy density was derived by the laser power and speed, which varied from the lowest 10.4 J/mm^3 to the highest 43.8 J/mm^3 . These samples were classified into three groups: low energy density group (samples 1, 2, 6, 7, 11, 12, 16, 17, 18, 21, 22, 23), high energy density group (samples 3, 4, 5), and optimal energy density group (samples 8, 9, 10, 13, 14, 15, 19, 20, 24, 25). For the low energy density group, the metal powders did not reach their melting point and thus did not sinter. For the high energy density group, fiber laser caused the formation of a carbonized layer on the plastic. During the carbonization, the carbonized plastic ash was dispersed in the air, and the metal powder did not adhere to the plastic substrate and was swept away by the recoater. In the optimal energy density group, a metal layer was apparently formed, but the metal powder partially permeated into the plastic substrate, forming a significant amount of porosity, resulting in a heterogeneous bonding between the metal and the plastic in the case without the mixture intermediate layer (Fig. 3b). Such results demonstrate that printing metal on plastic for bonding and lamination is challenging due to the thermal impact of the high-energy fiber laser on the plastic surface. It also suggests that metal printing should be performed at a low energy density to prevent the formation of a carbonized layer while ensuring sufficient energy density for metal sintering.

In this regard, we performed microstructural observations in the following sections to determine whether the plastic is damaged by laser heat accumulation, such as defective pore formation, carbonization, metal powder permeation, and low adhesion.

3.2. Cross-section comparison of plastic-metal interfaces

To investigate the interfacial microstructure between the plastic and metal, showing a cross-section of the region printed with Ti6Al4V on the Ti6Al4V-PVA mixture-deposited PETG surface, the backscattered images for phase contrasts were captured using high-resolution scanning electron microscopy (SEM), and elemental mapping was performed using energy-dispersive spectroscopy (EDS). For comparison with the mixture intermediate, the samples without the mixture intermediate were prepared as a reference.

As shown in Fig. 4a, the plastic-metal interface, especially the bottom PETG and the top Ti6Al4V, has significant epoxy regions corresponding to the non-elemental pore regions, showing an incomplete bond between the two materials. Note that the epoxy resin used to fix the sample for cross-sectional measurements was identified as the “carbon” element

because it has penetrated into the pores between the metal and the plastic. This explains that the metal layer did not bond with the plastic layer in a continual structure (see Fig. 4b). In addition, the laser heat accumulation not only caused severe thermal damage to the plastic but also partially melted the bottom plastic region, causing the metal powder to permeate into the plastic region and form contiguous pores in the plastic region. Consequently, the plastic was thermally decomposed at the interface, resulting in an incomplete plastic-metal bond.

In the case of the intermediate interface of Ti6Al4V and PVA mixture, as shown in Fig. 4c and d, thermal damage and metal permeation issues were significantly controlled. In contrast to the pure interface, the plastic region was almost maintained in its original state, indicating that the mixture intermediate protected the plastic layer from laser-induced thermal damage. Furthermore, the Ti6Al4V particles in the mixture partially formed a linkage structure with the metal layer, promoting effective bonding with the plastic-metal interface.

The side effects of metal powders permeating the plastic layer and causing interfacial pores should also be discussed. As shown in Fig. 4b and d, it was qualitatively confirmed that the presence of a mixture interface leads to a slight reduction in pore formation within the boundary region of the plastic-metal continuum. In particular, the mixture layer blocked the interfacial permeation of the metal powder and suppressed the amount of pore formation. This may be due to the uniformly dispersed metal powders within the interface acting as a buffering agent against direct permeation into the plastic. Although EDS provides a quick visual of the porosity (Fig. 4), it is worthwhile to measure the pore volume with a quantitative methodology. To obtain a more accurate analysis of the bonding area between the plastic and metal, X-ray microscopy (XRM) was used to measure the porosity based on 3D data in the following chapter.

3.3. Measurement of the porosity and metal powder permeation rate in the plastic zone through X-ray microscopy

The bonding interface between the plastic and metal was measured with a mixture volume of $1600 \times 1400 \times 300 \mu\text{m}^3$, which is entirely consistent with the EDS measurement range, as the region of interest (Fig. 5a and c). The absorption contrast tomography (ACT) of XRM was performed to measure the plastic-metal interface with high absorption contrast. The acquired tomography was analyzed for porosity with and without the mixture layer using PoroDict in GeoDict, a program capable of 3D image processing. As shown in Fig. 5b and e, the porosity level was obtained based on the accurate gray value thresholds of plastic and metal defined by a machine learning algorithm operating with the AI-Base function. The results of individual pore identification analyzed by PoroDict compare the amount of pore distribution in $2 \mu\text{m}$ increments of pore size (Fig. 6). The results revealed that the mixture intermediate

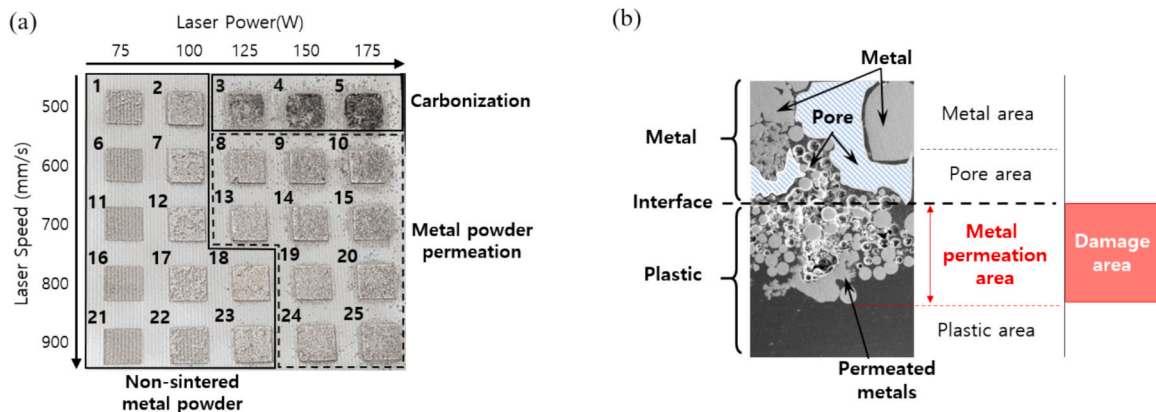


Fig. 3. Apparent metal layer formation depending on the processing conditions when printing metal on the plastic substrate. (a) 3D printed structures of Ti6Al4V according to laser power and speed variations and (b) the damage area at the Ti6Al4V-PETG interface in the case without the mixture intermediate layer.

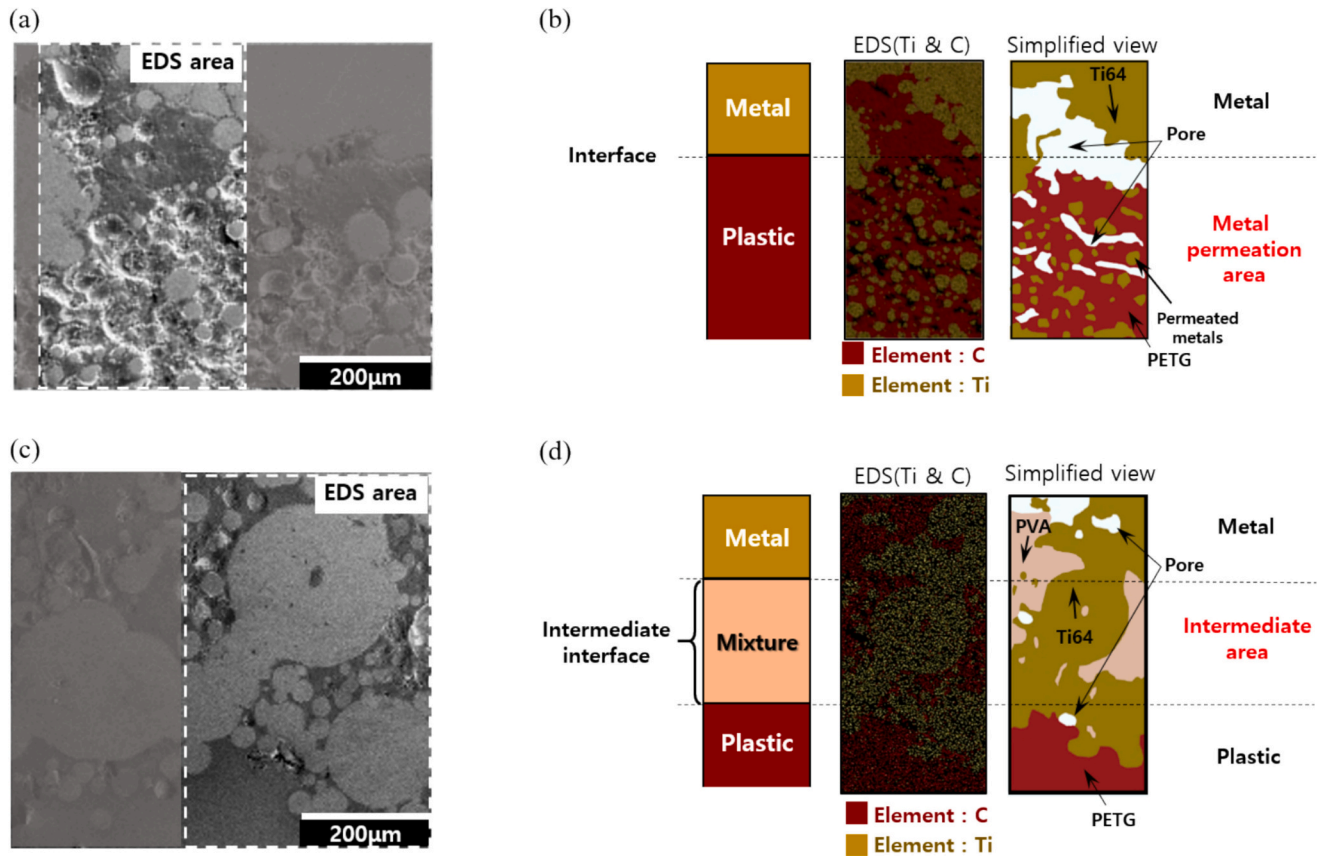


Fig. 4. SEM/EDS cross-sectional images of the interfaces. (a) Cross-section of PETG-Ti6Al4V and (b) the corresponding elemental mapping results of the interface structure. (c) Cross-section of the sample with the mixture intermediate and (d) the elemental mapping results of PETG-mixture-Ti6Al4V layers.

interface has a noticeably smaller amount of pore distribution at all pore size increments. As shown in Fig. 6a and b, the intermediate area contains rarely distributed pores with a diameter of 8 μm or less. On the other hand, the pure plastic-metal interface forms a metal-permeated zone and consists of massive pores larger than 10 μm. Particularly, it is noteworthy that the intermediate interface and metal permeation zone are morphologically similar but different in pore size distribution. The porosity analysis (Fig. 7a) revealed that the porosity at the mixture interface was almost half (5.2 %) compared to the sample without the mixture interface (11.1 %), with a standard deviation 6.5 times lower. This can be attributed to the high temperature and pressure of the fiber laser, which causes metal powders to be spattered, penetrating the plastic surface and increasing porosity due to physical impact, while the Ti-6Al-4V powders in the intermediate layer act as a barrier to suppress this penetration, thereby reducing the number of pores. These indicate that the mixture layer effectively controls the inevitable formation of pores, which cause bonding degradation. In Fig. 7b, the pore size distribution is represented by the cumulative percentages of 10 %, 50 %, and 90 % for the identified pores. According to the results, as the cumulative percentage of identified pores increases from 10 % to 90 %, the pore size of the pure interface increases significantly compared with that of the intermediate interface.

The “Percentile P90/P10” refers to a statistical ratio that compares the 90 % (P90) to the 10 % (P10) of a dataset; in the context of pore size distribution, the values for PETG-Ti6Al4V and PETG-mixture-Ti6Al4V are 10.2 and 7.3, respectively. This ratio indicates the range of pore sizes within these interfaces, where a higher ratio suggests a wider distribution of pore sizes, while a lower ratio indicates a more uniform pore size. This suggests that the mixture intermediate was more concentrated near the mean value (P50), resulting in a more homogeneous size distribution. That is, the pure plastic-metal interface shows a

wide distribution range in pore size, which means a heterogeneous plastic-metal interface. This variability in pore sizes could act as an unpredictable parameter for the printing process. In terms of fracture mechanics as well, a heterogeneous distribution of pore sizes can lead to unpredictable rapid fracture propagation by critical-size pores, whereas a homogeneous distribution of pore sizes can prevent catastrophic fracture, allowing reliable recognition of crack propagation.

The high-energy fiber laser causes metal powders to permeate significantly into the plastic body across the interface zone, contaminating the original purity of the plastic layer. The purity contamination of the plastic can impair the multi-functionality of the plastic-metal hybrids. Fig. 5c and f shows the metal permeation depth into the pure plastic region below the interface area of 300 μm depth limit. This analysis examines the protective effect of the mixture intermediate on the permeation of the fiber laser into the underlying plastic substrate. The plastic region of $1600 \times 1400 \times 600 \mu\text{m}^3$ was designated as the region of interest, and the metal powder was segmented by calculating the threshold value based on the Otsu algorithm thresholding method provided by GeoDict. In the absence of the mixture, approximately 16.6 % of the metal powder permeated across the 300 μm thick interface into the plastic zone. The metal permeation depth reached ~600 μm. In contrast, the mixture intermediate interface allowed only 1.2 % of the metal powder to permeate, and the metal penetration depth was <200 μm. The mixture intermediate hindered most of the metal powder penetration, as the Ti-6Al4V powders within the layer directly blocked the incoming metal powders. As a result, only a small amount of metal powder was observed in the upper region of the plastic. We expect that the purity preservation of the plastic by mixture intermediate can contribute to the harmonious integration of plastic-metal hybrids that simultaneously enable the multi-functionality of metals and plastics.

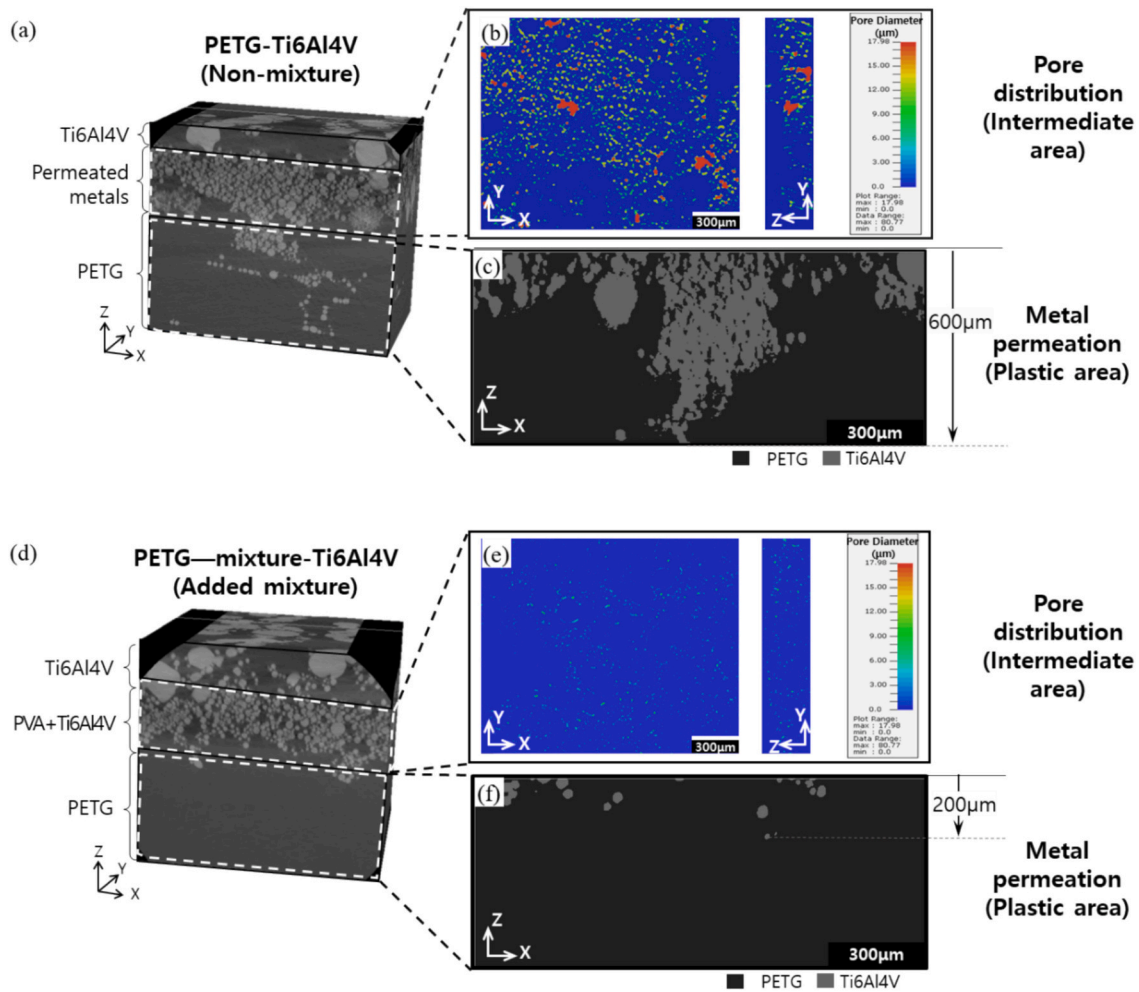


Fig. 5. XRM comparison of pore distribution between plastic-metal laminates and the penetration of metal powder into plastic. (a) CT image of the sample without the mixture. (b) Porosity in the metal permeation zone of PETG-Ti6Al4V without the mixture. (c) Metal permeation in the plastic area of PETG-Ti6Al4V without the mixture. (d) CT image of the sample with the mixture. (e) Porosity in the intermediate area of PETG-Ti6Al4V with the mixture. (f) Metal permeation in the plastic area of PETG-Ti6Al4V with the mixture.

3.4. Interfacial bonding strength depending on the presence of a mixture intermediate

To evaluate the effect of an intermediate layer on interfacial bonding strength, a customized metal jig was designed to measure the strength at the interface of dissimilar materials, as shown in Fig. 8a, which was reported in a previous study [2]. The test specimen was fabricated by depositing a Ti-6Al4V metal layer onto a 3D-printed rectangular PETG plastic substrate in the manner that allowed interlocking with a metal jig for fixation. As shown in Fig. 8b, a tensile-shear test was performed, and the interfacial bonding strength was calculated by dividing the maximum tensile force by the bonded area (10×10 mm). The measured bonding strengths are presented in Fig. 8c.

For the PETG-Ti6Al4V specimen without an intermediate layer, delamination occurred due to the formation of a carbonized layer at the interface. As a result, the specimen could not be held securely in the jig and detached prematurely, which marked 0 MPa in the Fig. 8c. In contrast, the PETG-Mixture-Ti6Al4V specimen with an intermediate layer demonstrated relatively sufficient adhesion between the two dissimilar materials to allow secure fixation in the jig having the maximum bonding strength of $1.56(\pm 0.11)$ MPa. Although bonding was achieved in the specimen with an intermediate layer, the measured bonding strength was lower than that reported in the previous study [2].

This result can be attributed not only to differences in material

combinations but also to the harsh nature of metal laser deposition on the plastic layer, which has led to weak interfacial bonding. Further in-depth investigations and experimental studies are necessary to enhance the bonding strength and optimize the process.

4. Conclusion

This research addressed the issues of imperfect bonding of plastic-metal and metal powder permeation during the printing of Ti6Al4V metals on PETG plastics using high-energy laser sintering by 3D printing process. To solve these problems, the pre-mixture of PVA and Ti6Al4V was deposited onto the PETG surface by sequential processing of CO₂ and fiber lasers. The interfacial porosity and metal permeability generated during the metal printing were analyzed using computed tomography, thus the bonding ability of the fiber laser for metal printing was demonstrated.

The analysis of cross-sections with and without the mixture intermediate showed a significant difference in the heat accumulation to PETG during metal printing. Ti6Al4V powder easily permeated the unprotected PETG surface, causing thermal damage and permanent degradation of the plastic top surface, resulting in an imperfect plastic-metal bond. In contrast, the application of a mixture intermediate of Ti6Al4V and PVA controlled the thermal damage and metal penetration, and partially connected the metal powder and the metal mixture

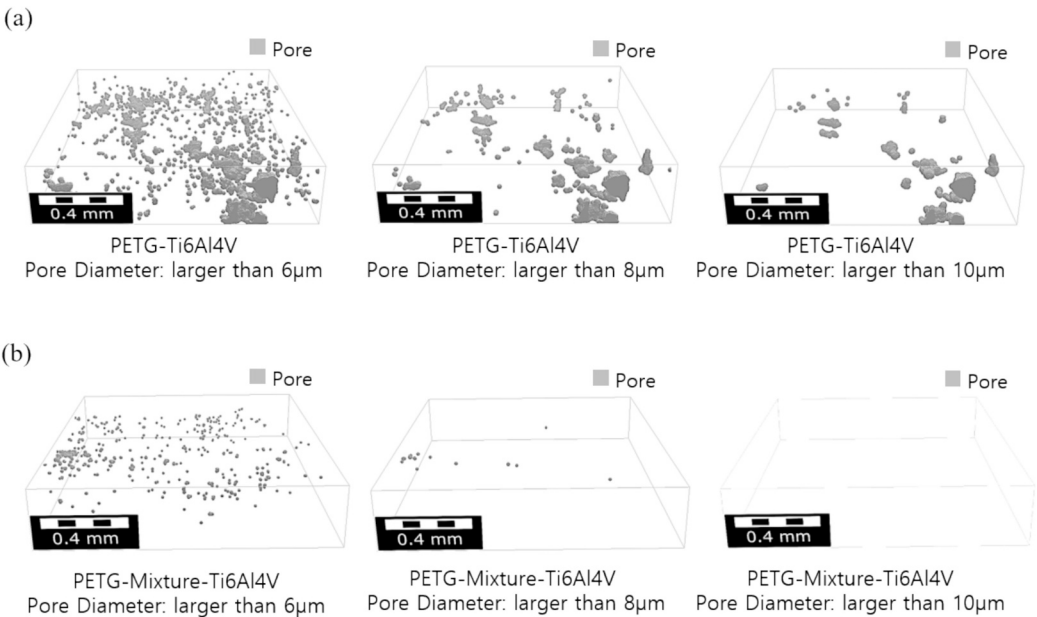


Fig. 6. Comparison of pores sizes between plastic-metal laminations. (a) Heterogeneous pore size distribution in the metal permeation zone of PETG-Ti6Al4V without the mixture. (b) Homogeneous pore size distribution in the intermediate area of PETG-Ti6Al4V with the mixture.

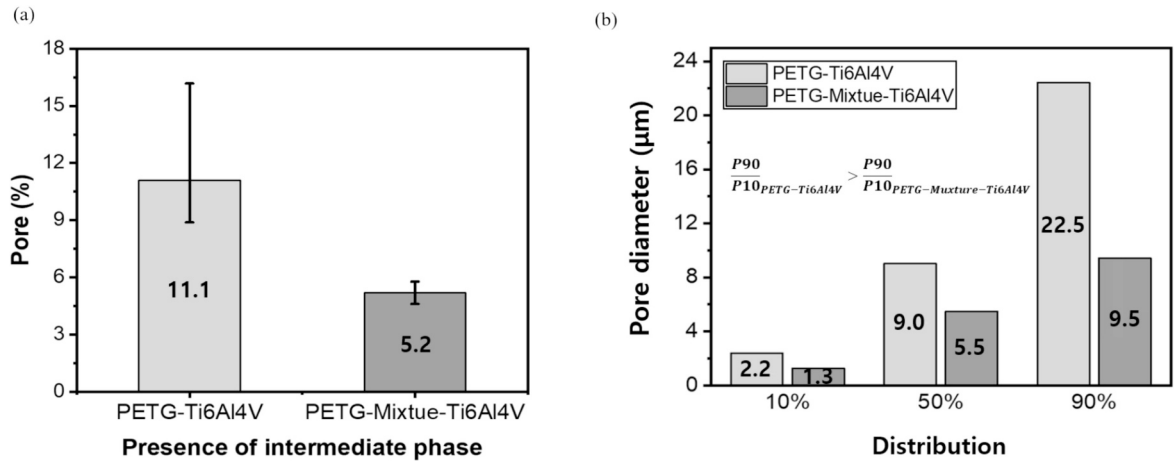


Fig. 7. (a) Internal porosities and standard deviation ranges based on XRM data. (b) Pore size distributions represented by the cumulative percentages of 10 %, 50 %, and 90 % for the identified pores.

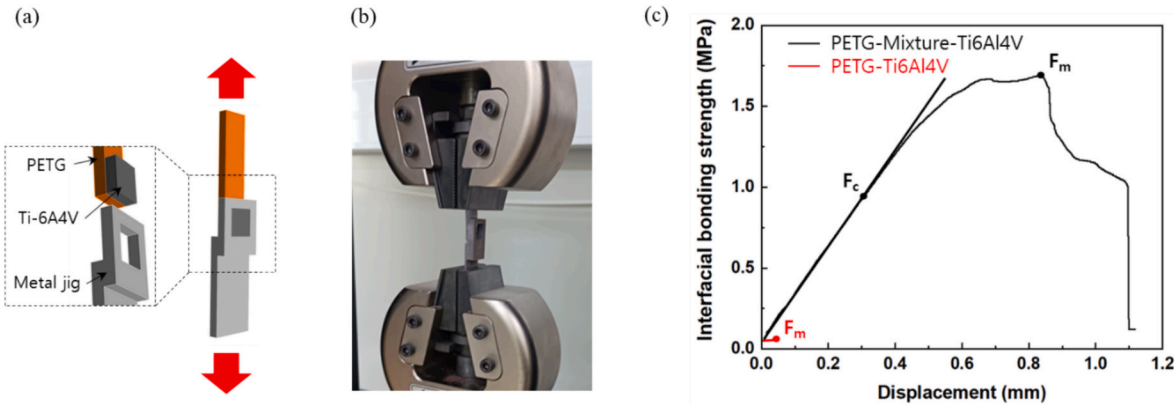


Fig. 8. Interfacial bonding strength evaluation at the plastic-metal bonding interface: (a) Schematic diagram for interfacial bonding strength measurement of PETG-Ti6Al4V and PETG-Mixture-Ti6Al4V specimens, (b) grip image of interfacial bonding test, (c) Tensile-shear bonding strength-displacement curve of the metal-on-plastic laminated specimen (F_c : critical point at which metal and plastic begin to detach, and F_m : maximum bonding strength).

intermediate, facilitating the bonding of the dissimilar plastic-metal.

The mixture intermediate significantly controlled the formation of critical-size pores that act as mechanical defects between plastic and metal. The pure interface without the mixture intermediate showed a heterogeneous distribution of pore sizes, forming a mechanically vulnerable interphase compared to the mixture intermediate with a homogeneous distribution of pore sizes.

The mixture intermediate prevents the interfacial permeation of metal powder into the plastic surface, thereby maintaining the original purity of the plastic and implementing a multifunctional plastic-metal hybrid structure.

This study is anticipated to be a key technology in plastic-to-metal bonding, enabling improved formability and minimized production processes for composite structures, with potential applications in advanced automotive and aerospace as well as medical devices.

CRedit authorship contribution statement

Geun Sik Shin: Writing – original draft, Visualization, Software, Methodology. **Eunho Cha:** Writing – review & editing. **Dae Young Kang:** Software, Investigation. **Jin Young Jung:** Software. **Hyeondeok Jeong:** Writing – review & editing. **Siwon Yu:** Writing – review & editing, Methodology. **Jun Yeon Hwang:** Writing – review & editing, Writing – original draft, Supervision, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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